35. The mean temperature of Monday to Wednesday was 37°C and of Tuesday to Thursday was 34°C. What was the temperature on Thursday ?

I. The temperature on Thursday was $\frac{4}{5}$th that of Monday.

II. The mean temperature of Monday and Thursday was 40.5°C.

III. The difference between the temperature on Monday and that on Thursday was 9°C.

(*a*) I and II only (*b*) II and III only
(*c*) Either I or II (*d*) Either I, II or III
(*e*) Any two of the three

36. In a cricket eleven, the average age of eleven players is 28 years. What is the age of the captain ?

I. The captain is eleven years older than the youngest player.

II. The average age of 10 players, other than the captain is 27.3 years.

III. Leaving aside the captain and the youngest player, the average ages of three groups of three players each are 25 years, 28 years and 30 years respectively.

(*a*) Any two of the three
(*b*) All I, II and III
(*c*) II only or I and III only
(*d*) II and III only
(*e*) None of these

Directions (*Question 37*): ***The given question is followed by three statements labelled I, II and III. You have to study the question and all the three statements given to decide whether any information provided in the statement(s) is/are redundant and can be dispensed with while answering the given question.***

37. What is the average salary of 15 employees?

I. Average salary of 7 clerical cadre (out of the 15 employees) is ₹ 8500.

II. Average salary of 5 officer cadre (out of the 15 employees) is ₹ 10000.

III. Average salary of the 3 sub-staff employees (out of the 15 employees) is ₹ 2500.

(*a*) None (*b*) Only I
(*c*) Only II (*d*) Only III
(*e*) Question cannot be answered even with information in all the three statements.

ANSWERS

1. (*c*)	**2.** (*c*)	**3.** (*e*)	**4.** (*e*)	**5.** (*d*)	**6.** (*d*)	**7.** (*d*)	**8.** (*b*)	**9.** (*b*)	**10.** (*d*)
11. (*e*)	**12.** (*e*)	**13.** (*e*)	**14.** (*c*)	**15.** (*e*)	**16.** (*e*)	**17.** (*e*)	**18.** (*a*)	**19.** (*d*)	**20.** (*b*)
21. (*a*)	**22.** (*d*)	**23.** (*e*)	**24.** (*a*)	**25.** (*e*)	**26.** (*e*)	**27.** (*b*)	**28.** (*e*)	**29.** (*e*)	**30.** (*c*)
31. (*d*)	**32.** (*a*)	**33.** (*d*)	**34.** (*b*)	**35.** (*c*)	**36.** (*c*)	**37.** (*a*)			

SOLUTIONS

1. I. Let their average salary be ₹ x. Then, Rajan's salary = ₹ $(x + 500)$.

$\therefore \frac{4000 + (x + 500)}{2} = x \Rightarrow 4500 + x = 2x \Rightarrow x = 4500.$

Sum of Rajan's and Sachin's salaries = ₹ (4500 × 2) = ₹ 9000.

$\therefore$ Rajan's salary = ₹ (9000 – 4000) = ₹ 5000.

So, I alone gives the answer.

II. Rajan's salary can be calculated from the given data as shown above.

So, II alone also gives the answer.

$\therefore$ Correct answer is (*c*).

2. From each one of I and II, we have: $r = \frac{x + y + z}{3}$.

i.e., r = average of x, y and z.

So, either I alone or II alone gives the answer.

$\therefore$ Correct answer is (*c*).

3. II. Let the marks obtained by Manick be x.

Then, 125% of $x = 80 \Rightarrow x = \left(\frac{80 \times 100}{125}\right) = 64.$

I. Average of marks of Nitin and Manick $= \frac{80 + 64}{2} = 72.$

$\therefore$ Arun's marks = 72.

So, both I and II together give the answer.

$\therefore$ Correct answer is (*e*).

4. I. $w + x + y + z = 25 \times 4 = 100.$ ...(*i*)

II. $w < 24$, $x < 24$, $y < 24$

$\Rightarrow \quad w + x + y < (24 \times 3)$

$\Rightarrow \quad w + x + y < 72$ ...(*ii*)

From (*i*) and (*ii*), we have: $z > 28$. So, z is the largest number.

Thus, both I and II together are needed.

$\therefore$ Correct answer is (*e*).